MATISSE

EDTconf
International Conference on Engineering Digital Twins

Monitoring Renewable Energy Communities Using Digital Twins via SAREF4ENER and a REST API

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cleanwatts digital

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Our Tasks in the MATISSE Project

- Proposing a model-based, federated digital twin for renewable energy communities (RECs) in cooperation with the Cleanwatts Digital (CWD) company in Portugal.
- Standardization of the project based on international/national standards.

Introduction

Introduction

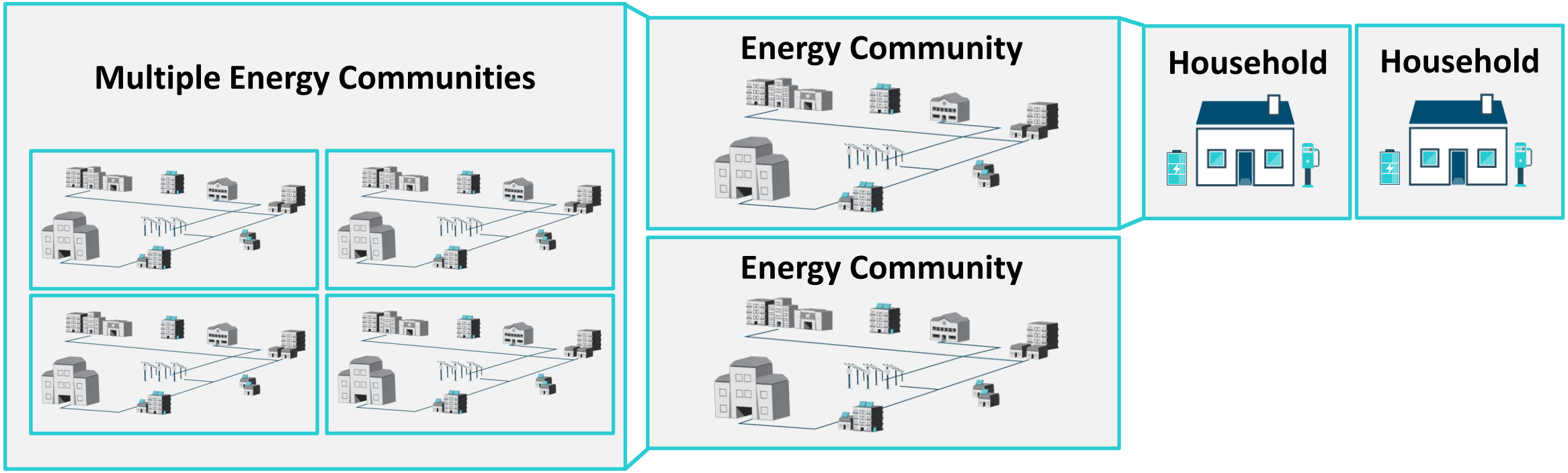
- RECs enable local prosumers and consumers to share energy resources, such as PV panels, for grid-level demand response.
- A DT is a synchronized software replica of a physical device, enabling monitoring, simulation, and control.
- In an REC, a DT allows operators to test decisions on a digital copy, minimizing risk and financial loss.
- Eclipse Ditto is our chosen open-source DT platform for representing each device as a Thing with named properties.

Introduction

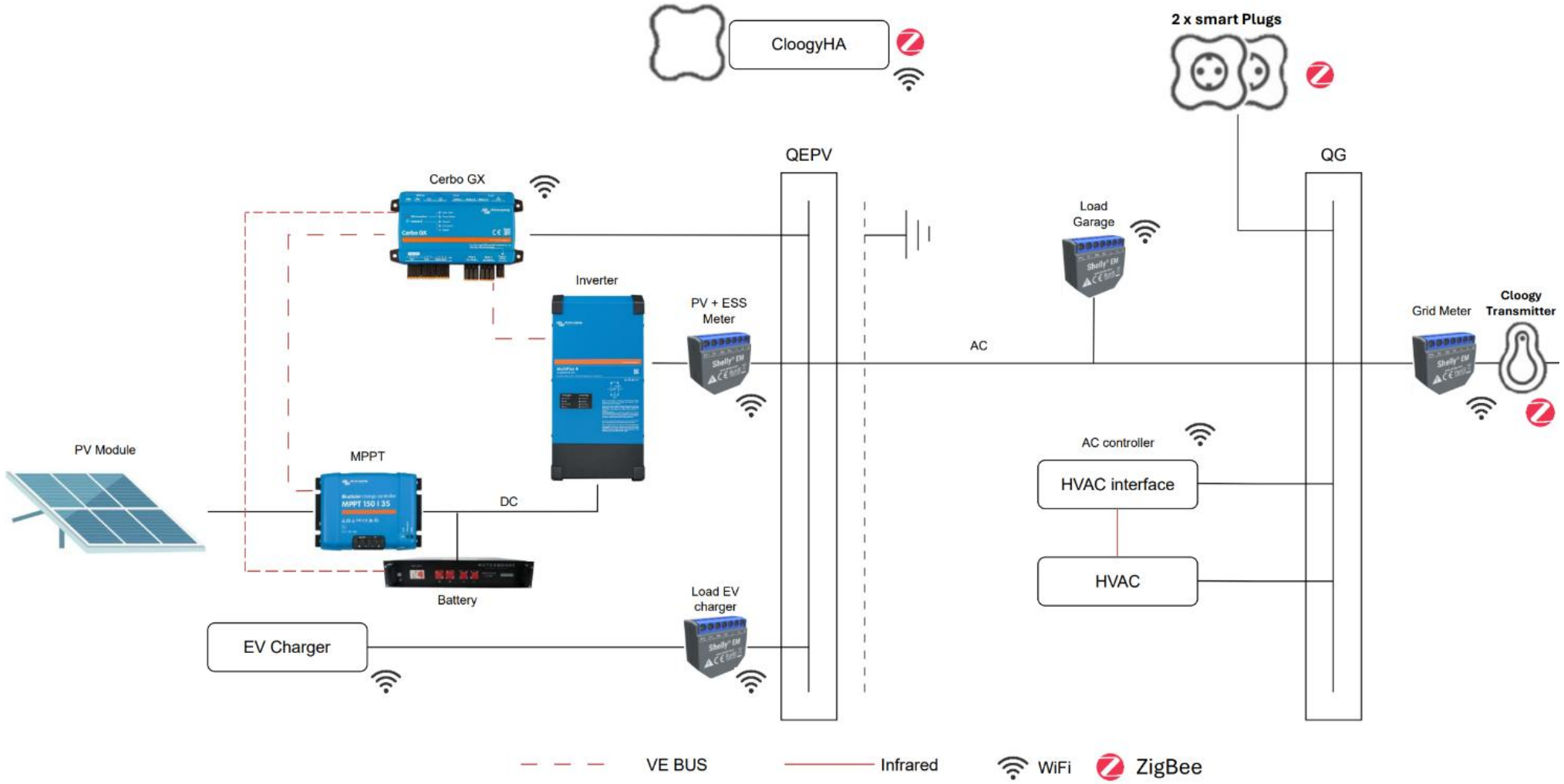
Two obstacles stand in the way:

- The system model is hard for third parties to interpret. Our target is the semantic interoperability by typing SysML model.
- The decisions a twin must support need real operational data, and acquiring it from a live community is not a generic ingestion task.

Introduction



Introduction



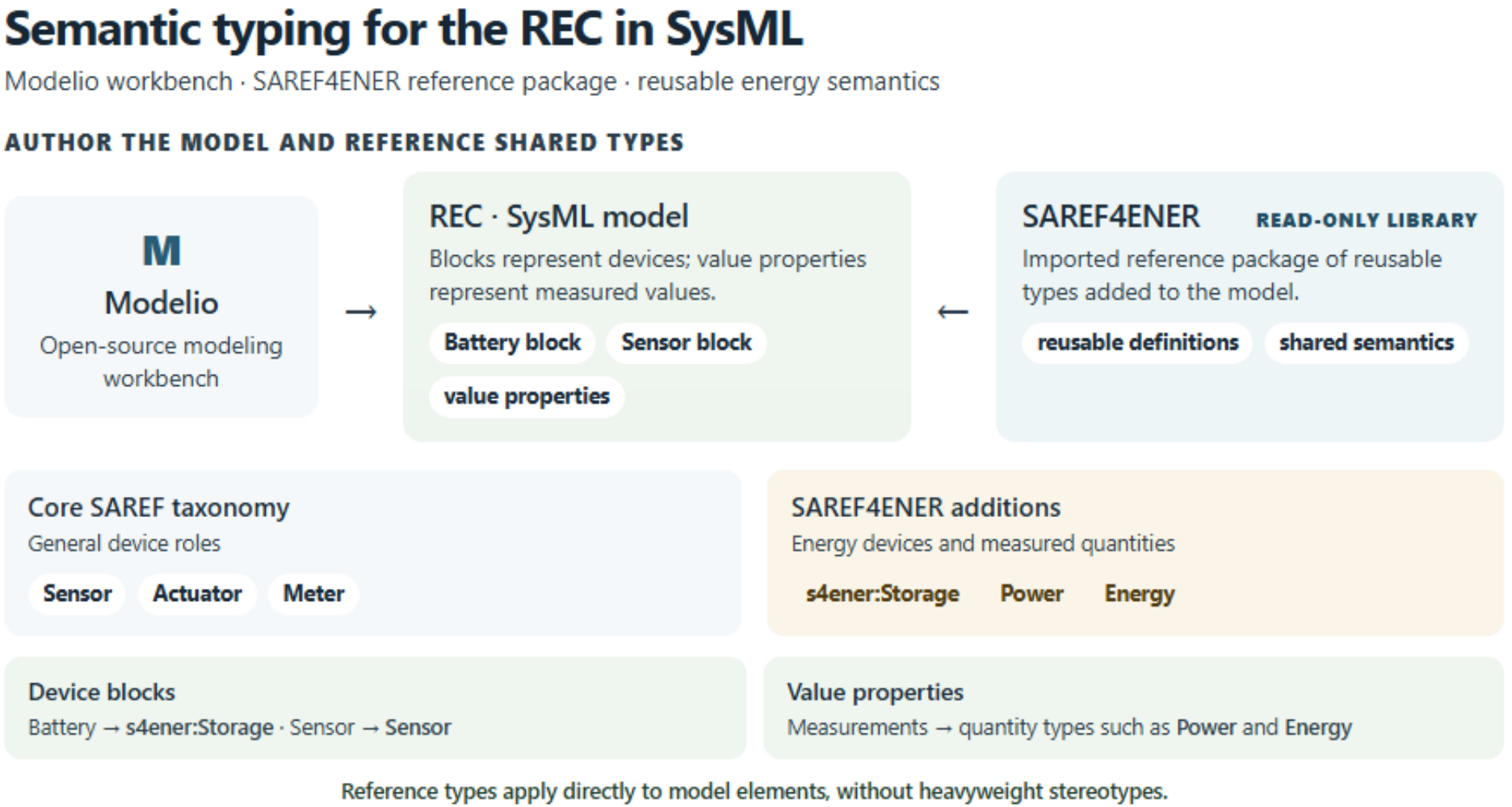
Contributions

Contributions

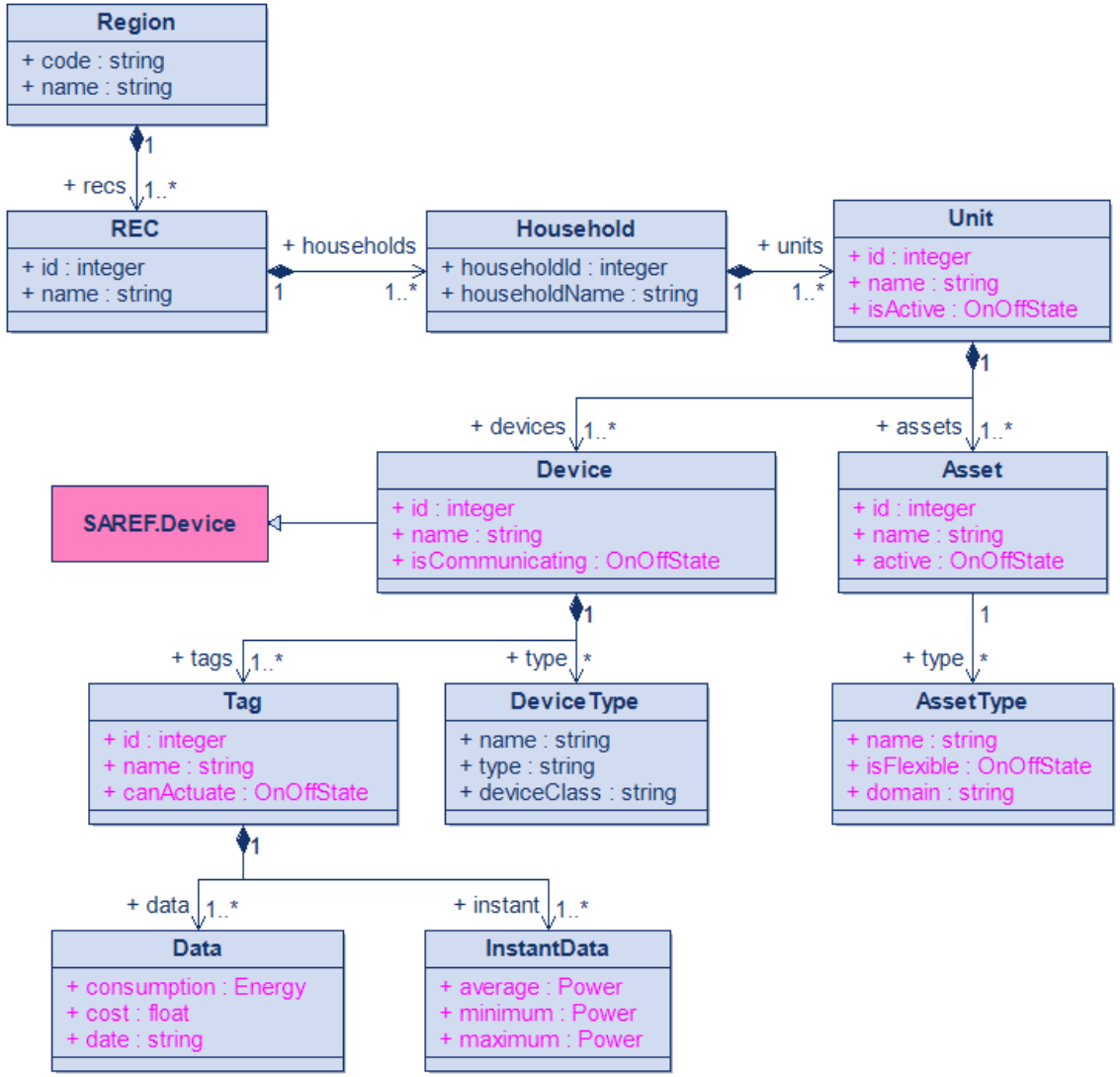
- A SAREF4ENER-grounded SysML modeling approach enhances REC system interoperability and understanding for third parties at data level.
- iEnergyRAPI acquires real-time data from CWD Living Lab via REST API for time-series storage.
- Experimental results show instantaneous device data, together with visualization through CWD Kisense.
- Lessons learned on closing design-to-runtime loop by reflecting runtime state into model.

Methodology

Methodology (SAREF4ENER Ontology)

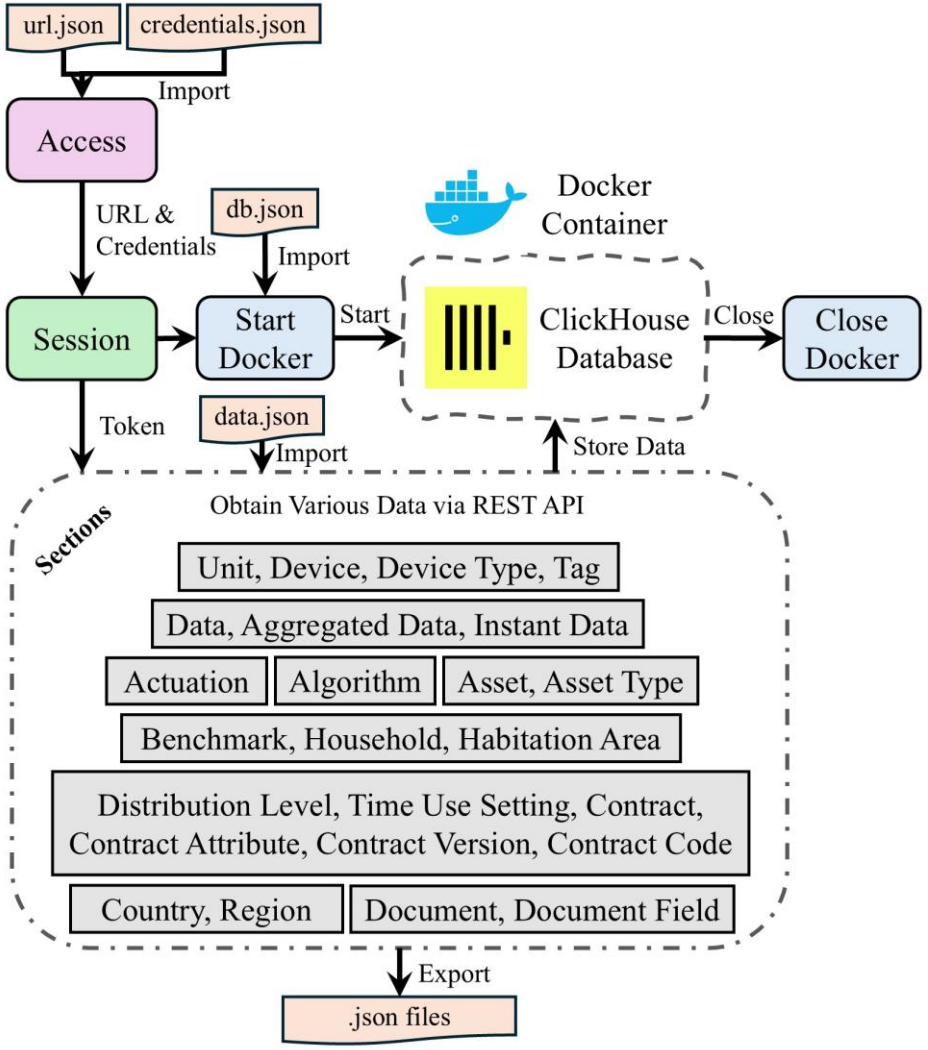


Methodology (SAREF4ENER Ontology)



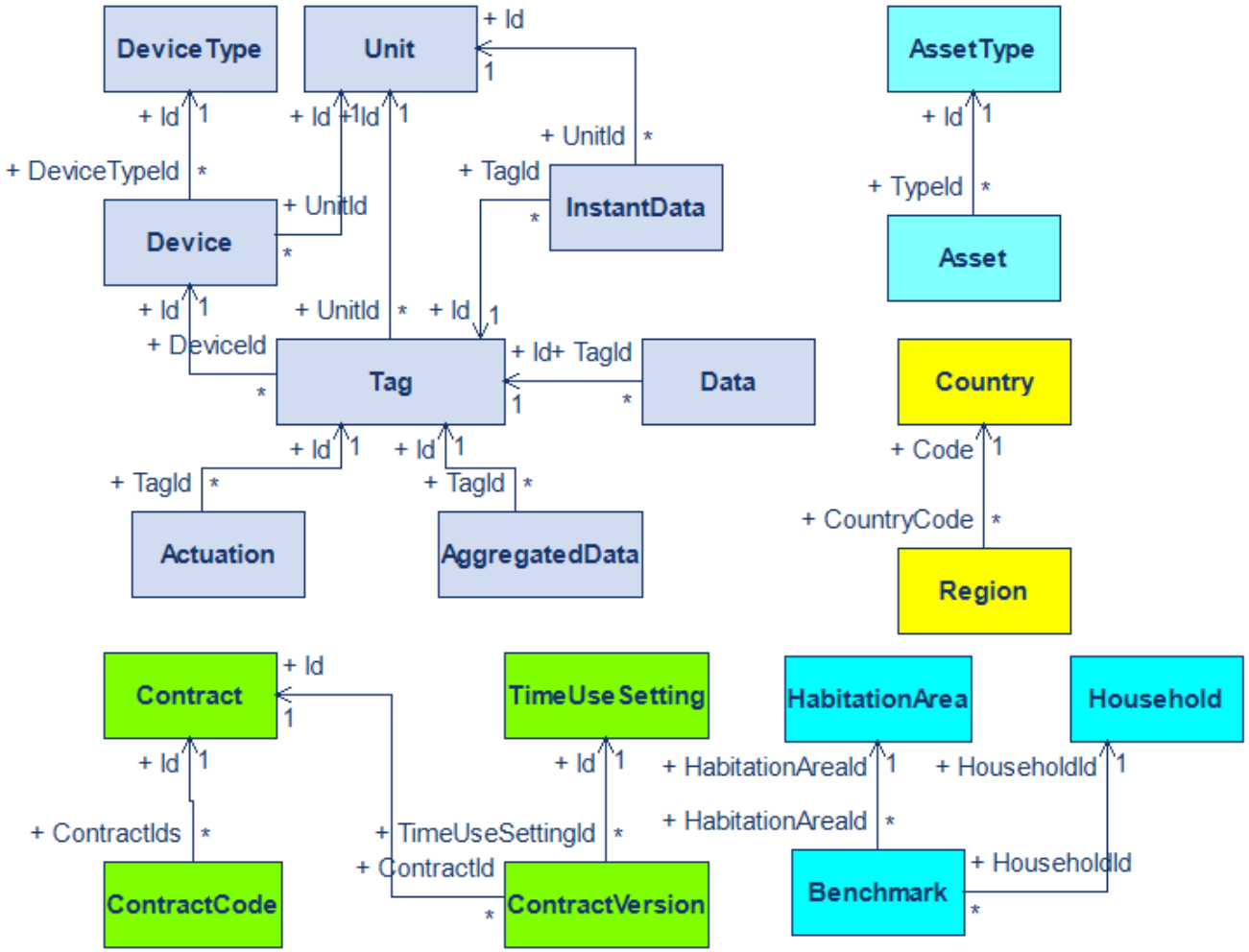
- Model as federation: region, community, household, and installations (units).
- Units group devices and energy assets; devices expose measurement points (tags) with SAREF types.
- Structural groupings utilize SAREF4ENER; interoperability through shared identifiers enabled.
- Distinction from iEnergyRAPI data schema; focuses on semantic view, not data storage.

Methodology (The iEnergyRAPI Tool)



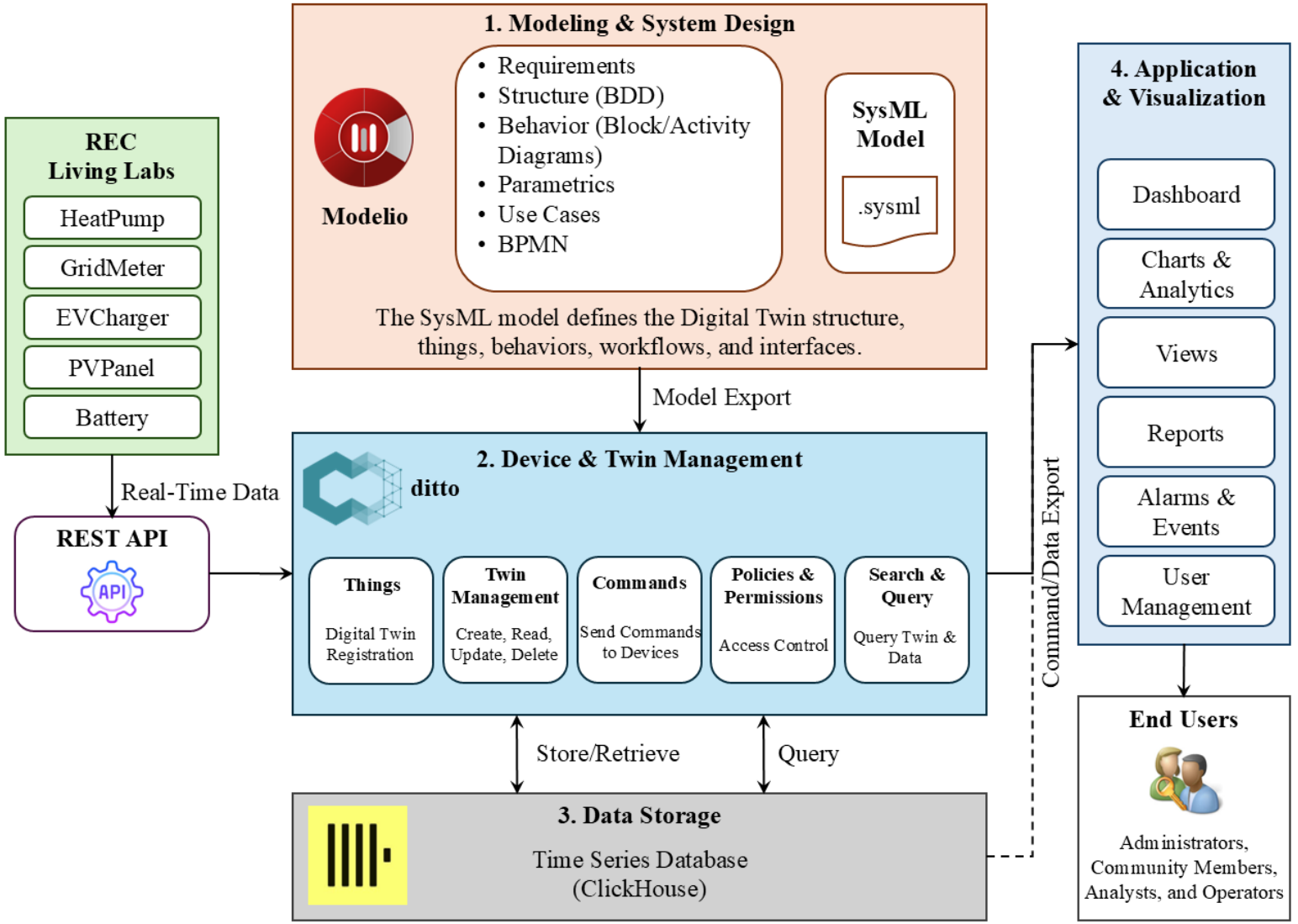
- iEnergyRAPI tool obtains data from CWD Living Lab via REST API.
- Session module connects, retrieves token for real-time data via GET requests.
- Responses converted to JSON for easy processing.
- Data stored in ClickHouse time-series database in Docker.
- Covers REC entities: units, devices, tags, time-series data, and energy assets.
- Normalizes missing values; exports each entity to JSON files.

Methodology (The iEnergyRAPI Tool)



- Schema's main entities organized by subdomain, with their associations and multiplicities.
- Device and measurement entities and the energy assets correspond to SAREF/SAREF4ENER concepts (saref:Device, saref:Measurement, saref:Command, and SAREF4ENER energy-device classes).
- Stored values match SAREF classes in semantic model.

Methodology (System Architecture)

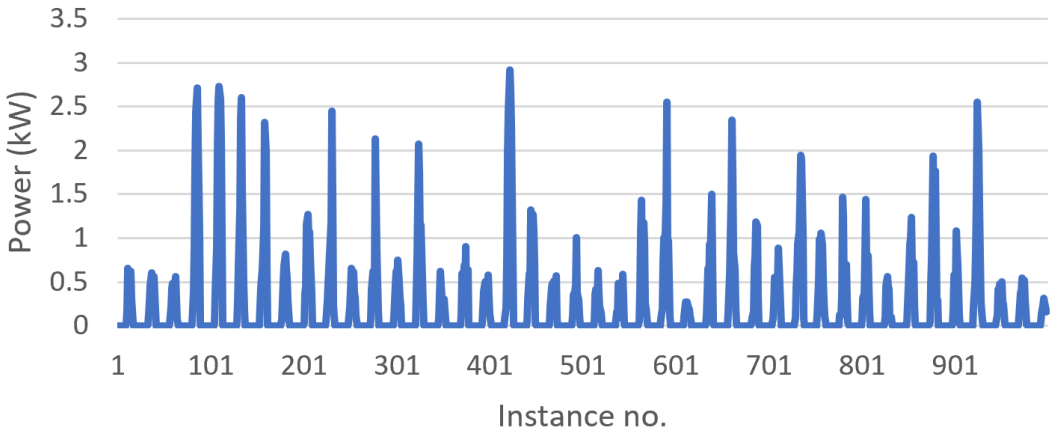


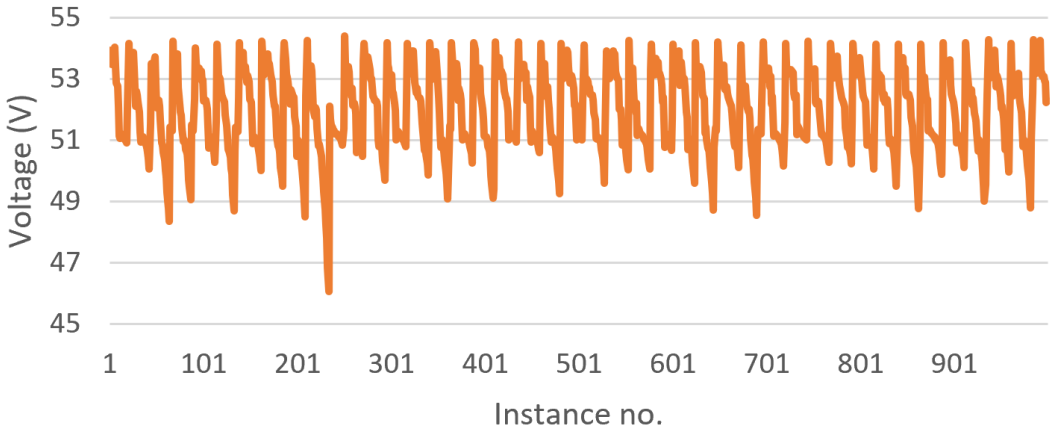
The role of Eclipse Ditto beyond future integration

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Experiments

Experiments



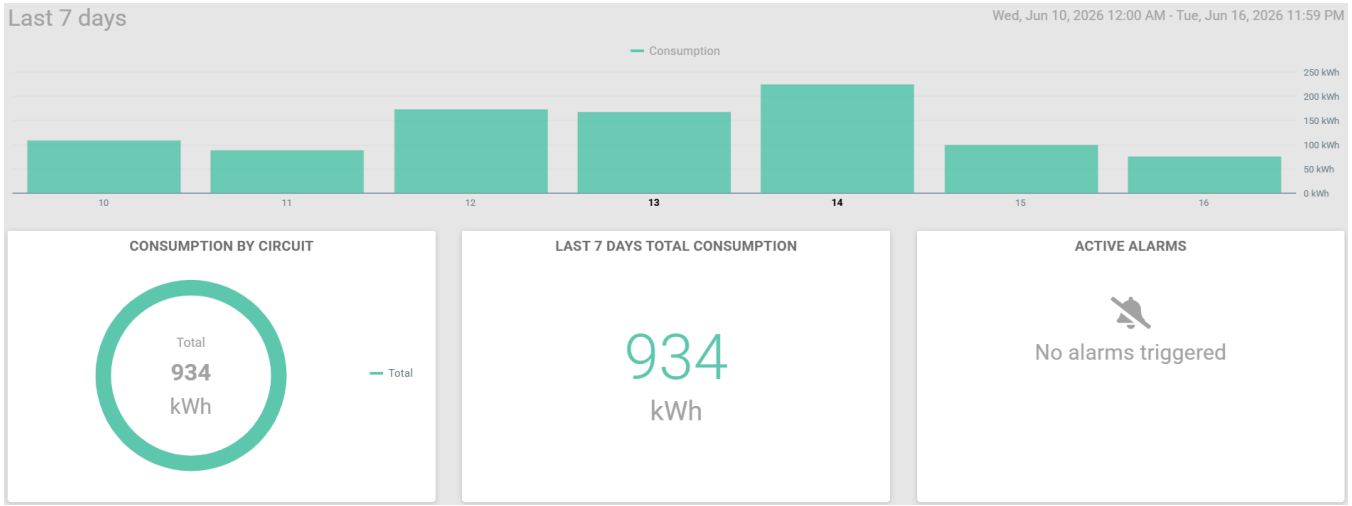


- Instant data from the PV meter at the Living Lab shows average power from [0, 3] kW over 1000 instances.
- Most values are within [0, 1.5] kW.
- Battery voltage data indicates a range of [45, 55] V with a minimum of 49 V in most instances.

Experiments

1	2337	FE3A7921EBFA46A38EC1C9AFD7B0BA6C	OMIE PT	106	443	0	1	2	0	FE3A7921EBFA46A38EC1C9AFD7B0BA6C	1		0	0	44	Other
2	5105	6B1B6746DE964D94BD006B6C03831564	Geral	225	1039	0	1	2	2	6B1B6746DE964D94BD006B6C03831564	1		0	0	shellyem-34945470F0C6:0	IpOrMc
3	5267	35255B7D8D1342269667470F4D5493D2	Geral	225	1069	0	1	2	4	35255B7D8D1342269667470F4D5493D2	1		0	0	shellyem-34945470EEB4:0	IpOrMc
4	7153	86E6624304844FEF877128FFDD06F047	Grid Meter	390	1499	0	1	2	0	86E6624304844FEF877128FFDD06F047	1		0	0	shellyproem50-08f9e0eac8e8:0	IpOrMc
5	7197	4373ECA4F51B4E97AE690DE648DF9262	Geral_new	390	1155	0	1	2	7	4373ECA4F51B4E97AE690DE648DF9262	1		0	0	192.168.1.65:0	IpOrMc
6	7301	7F0861EFF7F04B36BB8E6EE475FCF5E6	Victron	201	1496	0	0	1	4	7F0861EFF7F04B36BB8E6EE475FCF5E6	1		0	0	192.168.1.129:502:228	Modbu
7	7308	E8ADB73F045C4EF6BF506CD28CF4CCFB	Grid meter	225	1496	0	0	0	6	E8ADB73F045C4EF6BF506CD28CF4CCFB	1		0	0	192.168.1.50:0	IpOrMc
8	7601	BD2726A1F1114A99829BC9A2957ECCC1	Bateria Solis - Bateria	437	1551	0	1	2	4	BD2726A1F1114A99829BC9A2957ECCC1	6		0	0	192.168.68.155:502:1	Modbu
9	7611	54E92C8910A041E695C172DA44AE2D59	Shelly EM - Rede	225	1551	0	1	2	6	54E92C8910A041E695C172DA44AE2D59	6		0	0	shellyem-BCFF4DFD0781:0	IpOrMc
10	7613	BABA5F75415D49FB8F70DCD77129C73C	Shelly PM Mini G3 - Produção PV	441	1551	0	1	2	8	BABA5F75415D49FB8F70DCD77129C73C	6		0	0	ShellyPMMiniG3-ECDA3BC1723C:0	IpOrMc
11	7722	3D1F3B0588D645ABBBE3DD656097A7A9	PV meter	226	1496	0	1	2	9	3D1F3B0588D645ABBBE3DD656097A7A9	1	nan	0	0	192.168.1.51:0	IpOrMc
12	8423	FFEC51761539499CBA7A4351BF86FCBD	ERedes Total Shared	466	1703	0	0	0	0	FFEC51761539499CBA7A4351BF86FCBD	1	nan	0	0	5DF28A2678F5495AB3A4DA724E893CB3	Other
13	8424	FA8AE89894D946B6B5DAC1C1510FA6D3	ERedes Total Excess	465	1703	0	0	0	1	FA8AE89894D946B6B5DAC1C1510FA6D3	1	nan	0	0	6FAC8FF8613E4273B017AC10E7D78919	Other
14	8425	EE7670B3C62D46A897D17B0C58E358CF	Sofia Albuquerque REC Behind Meter	294	1701	0	0	1	0	EE7670B3C62D46A897D17B0C58E358CF	1	nan	0	0	1EE67B470F2243978A76BEC802710DC9	Other

Experiments



- Kisense platform provided by Cleanwatts Digital offers explorer, analysis, and control features.
- Last 7 days' data: total daily power consumption between 50-250 kWh.
- Total 7-day power consumption: 934 kWh.
- No active alarms reported.

Discussion

Discussion

The case shows three concern:

- SAREF4ENER enabled low-cost SysML interoperability by using reference packages, avoiding custom stereotypes.
- iEnergyRAPI continuously acquires and stores REC operational data for twins, ensuring data quality.
- Three lessons are learned: use ontologies, apply shared SAREF class, and isolate acquisition via a REST tool.

Discussion

The main gaps:

- Data not integrated into Eclipse Ditto from Living Lab.
- Interoperability only exercised structurally with SAREF classes.
- SAREF class appears as WoT @type annotation.

Future Work

Future Work

- Close design-to-runtime loop by integrating acquired data into Eclipse Ditto.
- Reflect runtime state onto the model and project SAREF4ENER profiles.

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Thank You!

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Find the tool's repository here.



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